

Northwind Traders: Global Sales & Operations Intelligence Report

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Tool: Microsoft Power BI

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1. Project Overview

This report explains the process and insights behind an interactive Power BI dashboard built for Northwind Traders, a global gourmet food supplier.

The goal of this project was simple. Give the executive team a clear view of what is actually happening across sales, products, shipping operations, and employee performance using real data, not assumptions.

The dashboard is structured into three focused pages, each answering key business questions around revenue trends, product performance, and operational efficiency.

You can access the full interactive dashboard here:

[View Interactive Dashboard](#)

2. Data Import and Preparation

The dataset consists of seven tables loaded into Power BI and connected through a relational model:

Table	Records	Purpose
Orders	830	Core transaction data
Order Details	2,155	Line level revenue and discount values
Products	77	Product catalogue and status
Customers	91	Customer location and profile
Employees	9	Sales team details and hierarchy
Categories	8	Product category groupings
Shippers	3	Shipping company details

Two key data quality observations came up during preparation.

First, shipped dates were missing for orders that were not yet shipped. These records were retained but excluded when calculating shipping days to avoid distortion.

Second, the "reportsTo" field in the employees table was blank for Andrew Fuller. This is expected because he is the head of the sales team and does not report to anyone.

Revenue was calculated using actual transaction values from the order details table. The formula used was unit price multiplied by quantity multiplied by one minus discount. This ensures revenue reflects what was truly paid, not just listed prices.

3. Dashboard Structure

The dashboard is divided into three main pages, each focusing on a different part of the business.

Page 1: Sales & Revenue Overview This page gives a top-level view of performance.

- i. Total Revenue is 1.27M
- ii. Year on Year Growth is 179.3 percent
- iii. Total Orders are 830
- iv. Average Order Value is 1.53K
- v. Average Freight Cost is 78.24

The monthly trend shows revenue movement across all years on a single timeline, with a note that 2015 only contains data from January to July.

A clear seasonal pattern appears. Sales are stronger in Q1 and Q4, while Q2 consistently drops across the years.

The USA and Germany are the top revenue markets, contributing 0.25M and 0.23M respectively.

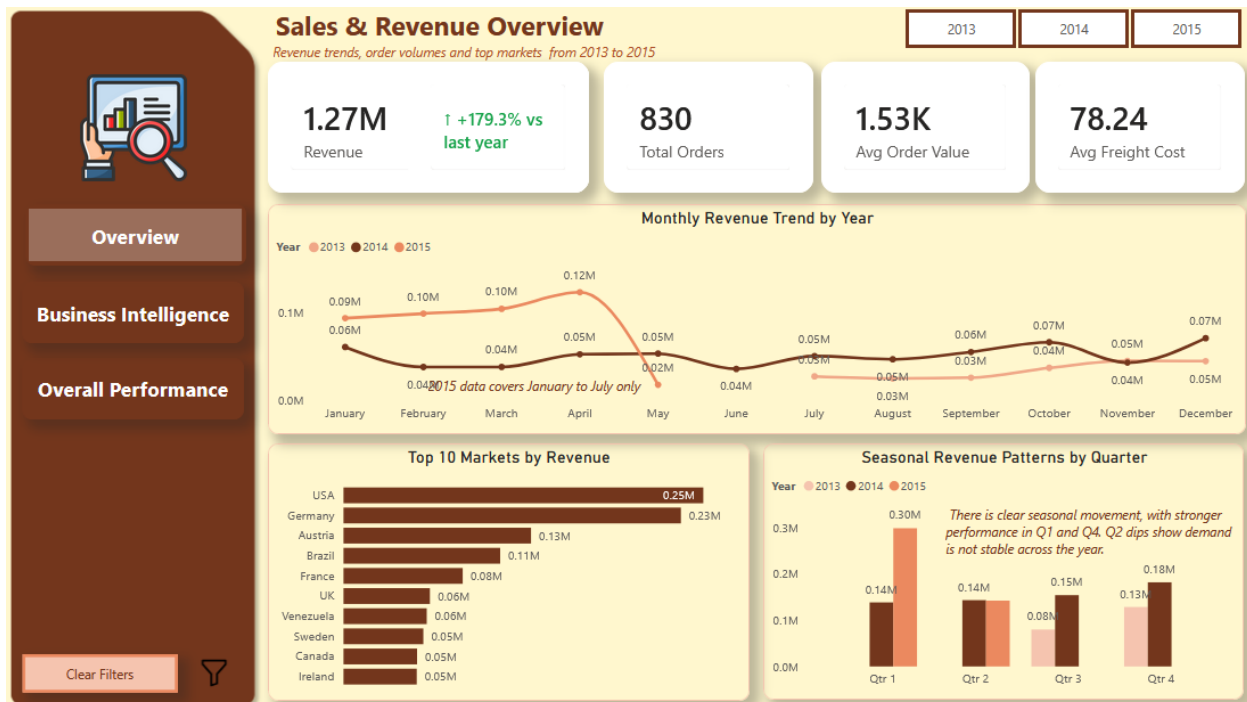


Figure 1: Sales and Revenue Overview Dashboard

Page 2: Product & Category Intelligence This page focuses on what is driving revenue at product level.

- i. Beverages is the highest performing category at 0.27M

- ii. Dairy Products follows at 0.23M
- iii. Côte de Blaye is the top product at 0.14M

One important finding here is that discontinued products still contribute 14.61 percent of total revenue. This is not a small number and suggests that some products may have been removed too early.

Discount analysis shows that average discounts across categories sit at 5.62 percent, with Meat and Poultry having the highest.

However, the scatter chart makes something clear. Higher discounts are not consistently leading to higher revenue. This means the business may be giving up margin without getting enough return in sales.

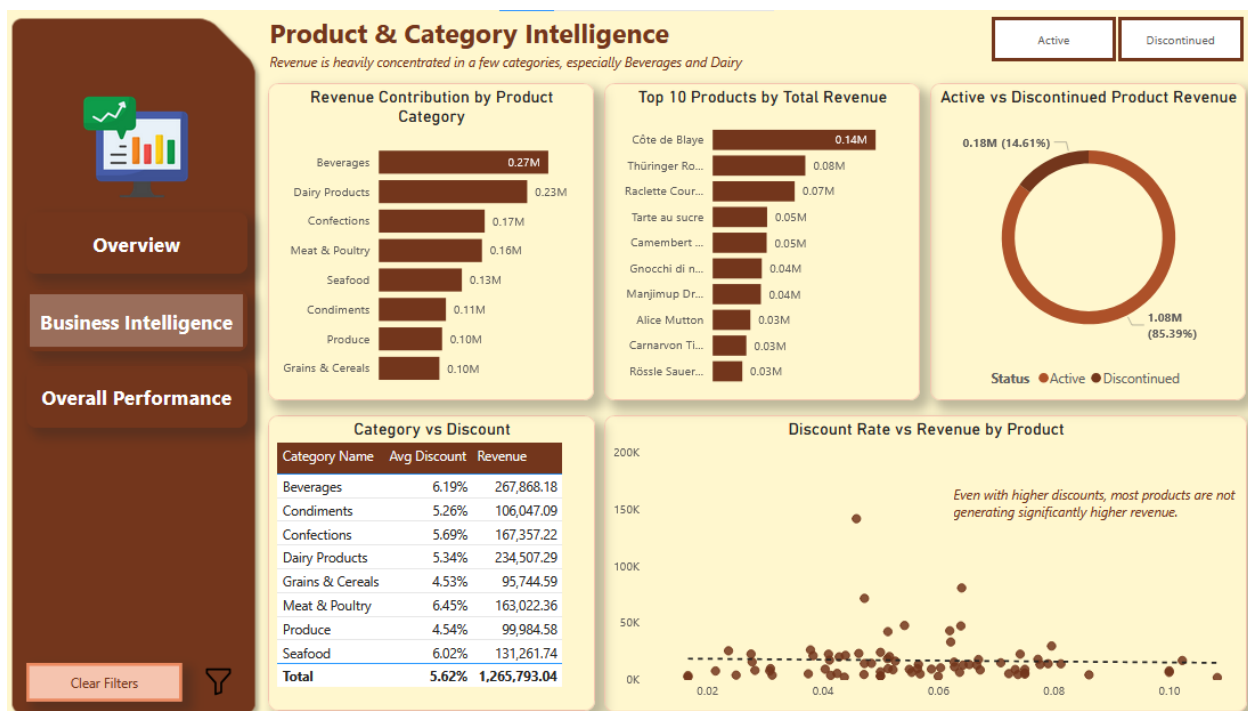


Figure 2: Business Intelligence Dashboard

Page 3: Operations & People Performance This page looks at execution, shipping and employee contribution.

- i. Margaret Peacock is the top performing employee with 0.23M in revenue
- ii. Steven Buchanan is the lowest at 0.07M

On shipping performance:

- i. United Package handled the most orders but also has the highest freight cost
- ii. Federal Shipping delivers the fastest, averaging 7 days

At country level:

- i. Austria has the highest average freight cost at 185
- ii. This is significantly higher than the USA at 113

This raises a business question. Is the revenue from Austria strong enough to justify that cost?

Total active customers across the dataset stand at 89.

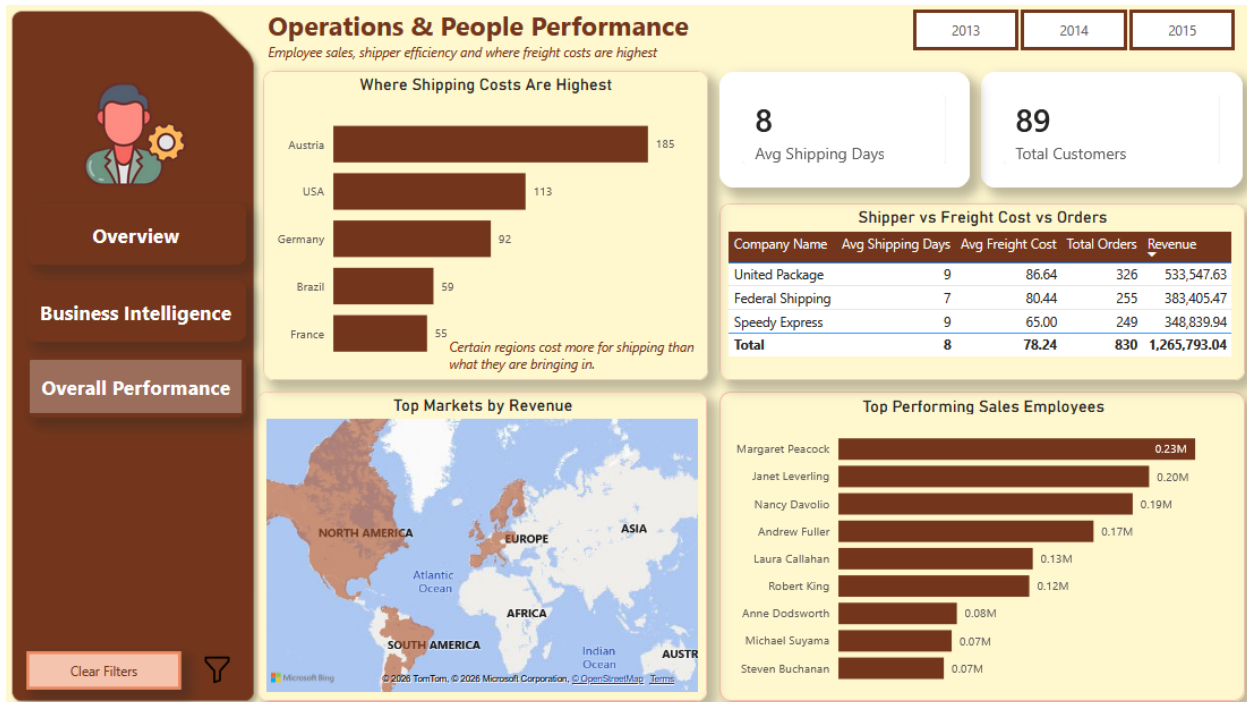


Figure 3: Operations Efficiency Dashboard

4. Key Insights

Three major things stand out from this analysis.

First, revenue growth looks strong at 179.3 percent, but this number needs careful interpretation. 2013 data start from July and 2015 ends in July, meaning only 2014 represents a complete full year. Leadership should factor this in when using the YOY metric for planning.

Second, discontinued products are still bringing in 14.61 percent of revenue. This suggests that product decisions may not be fully aligned with actual demand.

Third, discounting is not working as expected. Products with higher discounts are not consistently generating higher revenue, which means margin is being lost without clear benefit.

5. Filters and Interactivity

The dashboard was designed to be fully interactive.

- i. Year slicer allows users to filter performance by time
- ii. Product status slicer allows switching between active and discontinued products
- iii. Employee slicer helps analyse individual performance
- iv. Shipping company slicer allows comparison across shippers

All visuals support cross filtering, so selecting any data point updates the rest of the page automatically. This allows users to explore insights without needing separate reports.

6. Tools and Measures Used

All calculations were built using DAX in Power BI.

- i. Revenue was calculated using unit price, quantity and discount
- ii. Year on Year Revenue compares current performance with the same period last year
- iii. Average Shipping Days excludes orders without shipped dates
- iv. Average Freight Cost was calculated directly from the orders table

7. Recommendations

Based on the analysis, two actions stand out as priority for the business.

I. **Review and Reintroduce High Performing Discontinued Products**

The Product and Category team should carry out a quick audit of discontinued products that are still generating revenue. Products contributing to the 14.61 percent revenue share should be reviewed for possible relisting or replacement with similar alternatives.

Expected outcome within 60 to 90 days is an increase in category level revenue without additional acquisition cost, since demand for these products already exists.

II. **Restructure Discount Strategy Across Categories**

The Growth and Commercial teams should review how discounts are being applied, especially in categories like Meat and Poultry where discounts are highest but not translating into stronger revenue performance. Instead of blanket discounts, the business should test targeted or performance based discounts on specific products.

Expected outcome within 60 to 90 days is improved profit margin and better alignment between discount spend and actual revenue growth.